Practical 5

In an operating system three CPU-intensive processes are ready for execution, which require 10ns,20ns and 30ns and arrival at times Qns,2ns and 6ns, respectively. Write a program td calculate the total number of context switches needed if the operating system implements a shortest job first (preemptive) scheduling algorithm. Also calcutate the average time for which the processes have to wait before getting the CPU

Programiz
C Online Compiler

main.c

```c
#include <stdio.h>

int main() {
    int arrival[3] = {0,2,6};
    int burst[3] = {10,20,30};
    int start[3] = {0,10,30};
    int wait[3];
    int i;
    float avg=0;

    for(i=0;i<3;i++){
        wait[i] = start[i] - arrival[i];
        avg += wait[i];
    }

    avg = avg/3;

    printf("Waiting times:\n");
    for(i=0;i<3;i++){
        printf("P%d = %d ns\n", i+1, wait[i]);
    }

    printf("Average waiting time = %.2f ns\n", avg);
    printf("Total context switches = 2\n");

    return 0;
}
```

Output

```
Waiting times:
P1 = 0 ns
P2 = 8 ns
P3 = 24 ns
Average waiting time = 10.67 ns
Total context switches = 2


=== Code Execution Successful ===
```